

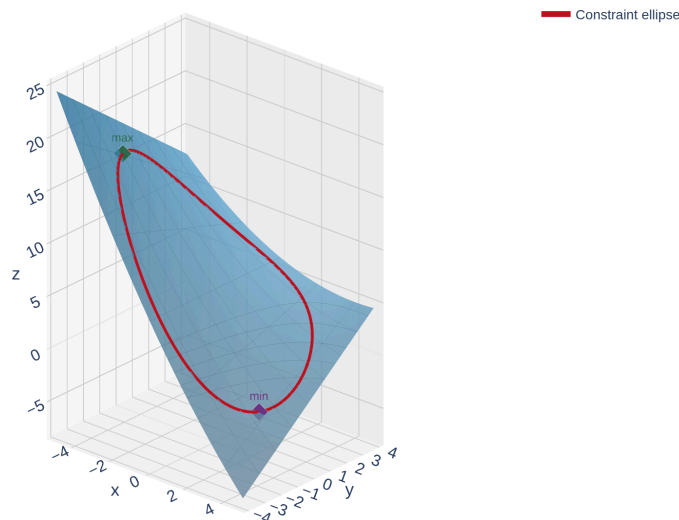
Why Constrained Optimization?

Lagrange multipliers give us an elegant, systematic method for optimizing a function **subject to a constraint** — no parameterization needed.

A Motivating Example

Example 1

Find the absolute maximum and minimum of $f(x, y) = 6 - 2x + 0.1x^2 + 0.3xy$ on the ellipse $0.2x^2 + 0.3y^2 = 3.6$.



The surface $z = f(x, y)$ restricted to the ellipse | [Interactive version](#)

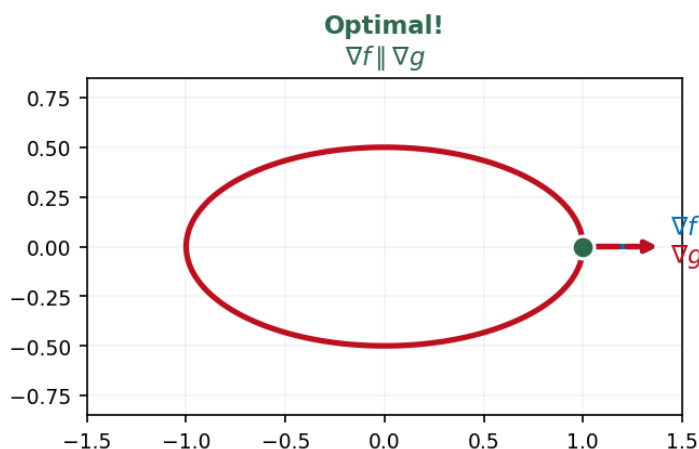
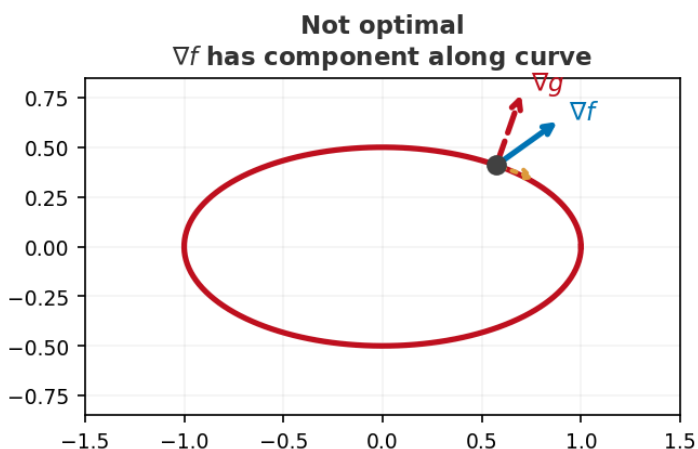
Approach 1: Parameterize the Constraint

The ellipse $0.2x^2 + 0.3y^2 = 3.6$ can be parameterized as:

$$x = \sqrt{18} \cos t, \quad y = \sqrt{12} \sin t, \quad 0 \leq t < 2\pi$$

Parameterization **works** but has drawbacks: finding a parameterization can be difficult, and the resulting function can be messy. We need a better way.

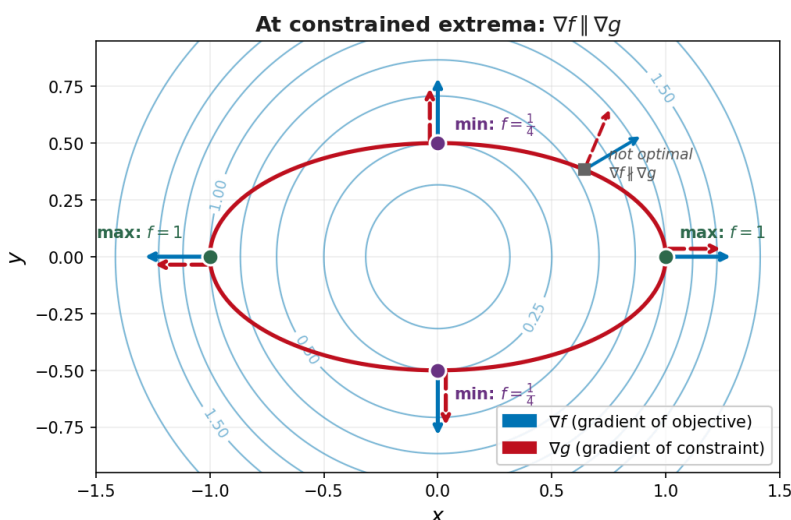
The Geometry Behind Lagrange Multipliers



At a constrained extremum, ∇f and ∇g point in the **same (or opposite) direction**. The scalar λ is called the **Lagrange multiplier**.

See It on a Contour Map

Here's the geometry in action: level curves of $f(x, y) = x^2 + y^2$ (blue circles) with the constraint ellipse $x^2 + 4y^2 = 1$ (red).



The Method of Lagrange Multipliers

Method of Lagrange Multipliers

To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = k$ (assuming these extreme values exist and $\nabla g \neq \vec{0}$ on the curve $g = k$):

1. Find all values of x , y , and λ such that

$$\nabla f(x, y) = \lambda \nabla g(x, y) \quad \text{and} \quad g(x, y) = k$$

1. Evaluate f at all the points (x, y) found in step 1. The largest value is the **maximum**; the smallest is the **minimum**.

You often don't need the actual value of λ — it can be eliminated from the system to find x and y .

Quick Check: Set Up a Lagrange System

Suppose you want to minimize $f(x, y) = x^2 + 2y^2$ subject to $x + y = 4$.

Write the three equations you'd need to solve (but don't solve them).

Identify f and g , compute ∇f and ∇g , then write $f_x = \lambda g_x$, $f_y = \lambda g_y$, and $g = k$.

Example 2: Closest and Farthest from the Origin

Example 2

Find the maximum and minimum values of $f(x, y) = x^2 + y^2$ on the ellipse $x^2 + 4y^2 = 1$.

Set up: $f(x, y) = x^2 + y^2$, $g(x, y) = x^2 + 4y^2 = 1$.

Write $\nabla f = \lambda \nabla g$ and $g = 1$. You'll get a system with cases — set factors to zero.

Your Turn: Lagrange Practice

Example 3

Find the maximum and minimum values of $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$.

Hint: Set up $\nabla f = \lambda \nabla g$ where $g(x, y) = x^2 + y^2$.

Set up the three equations, then solve for x , y , and λ .

Three Variables: $f(x, y, z)$ with Constraint $g(x, y, z) = k$

Same idea, one more variable. The system is bigger but the strategy is the same: use the gradient equations to relate the variables, then plug into the constraint.

Quick Check: Set Up a 3-Variable Lagrange System

Minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + 2y + 3z = 6$.
Write the four equations you'd need to solve (but don't solve them).

Identify f and g , compute ∇f and ∇g , then write $f_x = \lambda g_x$, $f_y = \lambda g_y$, $f_z = \lambda g_z$, and $g = k$.

Example 4: Minimum Surface Area Box

Example 4
Find the dimensions of a rectangular box with volume 1000 cm^3 that has the **smallest** surface area.

Set up:

- Surface area: $f(x, y, z) = 2xy + 2xz + 2yz \leftarrow$ minimize this
- Volume constraint: $g(x, y, z) = xyz = 1000$

Compute gradients, write the 4 equations, and look for symmetry!

When to Use Lagrange Multipliers

Step 1: What kind of region?

Region Type	What to Do
Open region (no boundary)	Find critical points of f ; use Second Derivatives Test (14.7)
Closed & bounded region	Check interior critical points + optimize on the boundary separately
Boundary Method	Best When
Parameterize the curve	Boundary has a simple parameterization (line, circle)
Lagrange multipliers	Constraint is hard to parameterize, or you have multiple constraints

Lagrange multipliers don't replace the critical-point method — they handle the **boundary step** that the critical-point method can't.

Homework

Section 14.8: 1-13 odd, 45